

Supply Chain Performance Analysis Report

Executive Summary

This report presents an end-to-end analysis of supply chain delivery performance. A total of **172,765 orders** were analyzed. The findings reveal that **54.71% of orders are delayed**, while only **45.29% are delivered on time**. These delays result in a significant financial impact, with an estimated **\$2.1M loss due to delays**, despite generating a total profit of **\$7.5M**. The analysis identifies operational inefficiencies and key delay drivers across regions, shipping modes, and time periods.

Dataset Overview

The dataset contains transactional order-level data including shipping timelines, customer segments, product details, and profitability metrics. Data cleaning was performed to remove redundant, irrelevant, and low-variance columns, ensuring high-quality analysis.

Key Performance Indicators (KPIs)

- Total Orders: 172,765
- Late Deliveries: 94,523
- On-Time Delivery Rate: 45.29%
- Late Delivery Rate: 54.71%
- Total Profit: \$7.5M
- Loss due to Delays: \$2.1M

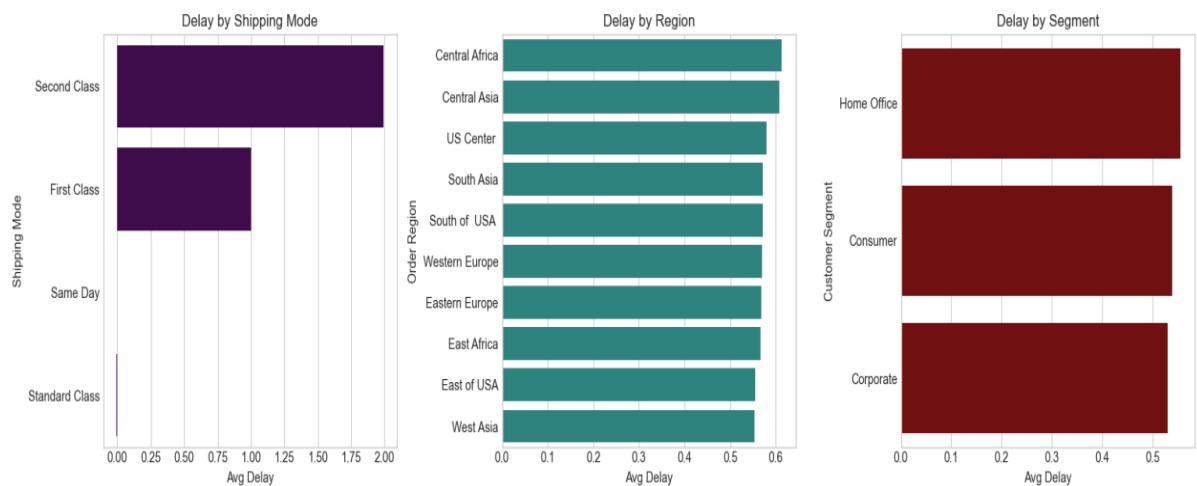
Bottleneck Analysis

The analysis highlights that specific shipping modes and regions contribute disproportionately to delivery delays. Certain categories consistently show higher delay percentages, indicating operational bottlenecks in logistics and fulfillment.



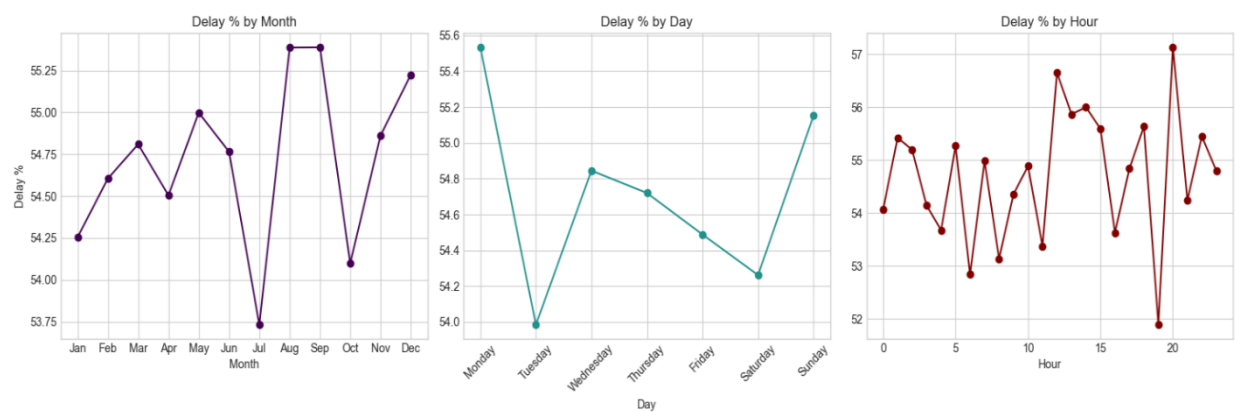
Root Cause Analysis

Key factors contributing to delays include inefficient shipping modes, longer order processing times, and regional logistics challenges. Orders with higher processing times show a strong correlation with delayed deliveries.



Time-Based Analysis

Delay patterns vary significantly across months, days, and hours. Certain time periods experience higher delays, indicating workload imbalance and operational inefficiencies during peak times.



Business Recommendations

- Optimize high-delay shipping modes to improve delivery speed
- Reduce order processing time through automation
- Focus on high-delay regions for logistics improvement
- Implement monitoring during peak delay hours
- Use predictive analytics to identify at-risk orders

Conclusion

The supply chain currently faces significant delay challenges affecting both profitability and customer experience. By addressing identified bottlenecks and optimizing operations, the business can improve delivery performance and reduce losses.

